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Controller design for high-speed, ultra-precision positioning of a linear motion stage on a vibrating machine base stage control on a vibrating base

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ABSTRACT

This paper presents a high-speed, ultra-precision point-to-point position control system design. It addresses precision stages with unknown characteristics, affected by machine base vibration and system effectiveness. The design is based on the nominal characteristic trajectory following (NCTF) control method, which can provide ultra-precision position control performance for the precision stages without accurate dynamic models. However, conventional NCTF control systems neither provide sufficient vibration suppression nor exhibit suitable vibration suppression characteristics. The position control performance is deteriorated by the machine base vibration. To overcome this problem, a procedure for incorporating a bandpass filter and derivative compensator in the NCTF control system is proposed and designed. These compensators exhibit high vibration suppression ability when combined, rather than when used individually. Additionally, they can be easily designed without any accurate dynamic model. The effectiveness of the compensator combination was first investigated using a linear model and subsequently experimentally verified. Using the proposed procedure, a control system was designed for the precision stage with friction characteristics on the vibrating machine base. The designed control system suppresses the residual vibration immediately after the target value becomes constant, and the error converges below 50 nm within 80 ms.

1. Introduction

Position accuracy and motion speed are the most basic characteristics of precision-motion mechanisms. In manufacturing systems, the position accuracy of the motion mechanism determines the quality of the product and influences production efficiency. In recent years, high product quality and production efficiency have become increasingly important. Therefore, improving these characteristics is desirable [1,2].

In addition to friction and backlash, mechanical vibration is a typical factor that reduces position accuracy. High acceleration/deceleration for high-speed motion generates a large reaction force, which increases vibrations and causes new problems. In addition to the low-stiffness elements in the precision motion stage, the machine base vibrates, producing nonnegligible residual vibrations. Heavy and high-stiffness machine bases and active vibration isolators [3–5] are effective in suppressing vibrations. However, to reduce costs and ease restrictions on the installation environment, it is necessary to reduce the weight of manufacturing machines [6,7].

Performance improvement by a suitable control method does not require hardware changes. Furthermore, it can be conveniently used and applied to many machines, including existing machines. The important aspect of position control is to precisely control the relative position with respect to the reference position and to rapidly suppress the vibration of the relative position. In this study, the reference position has been fixed to the machine base. Various control methods have been applied to feed drive systems [8]. Traditional controllers such as P-PI and PID controllers can be tuned easily and used widely in the feed drive system [9,10]. However, it is difficult for traditional controllers to suppress the machine's relative vibration, including the machine base [11]. To overcome this problem, the design methods of state feedback (FB) control systems [12] and two-degree-of-freedom (two-DOF) control systems [6,7,13–16] have been reported. Their control system performances have also been examined. These studies have shown that control methods provide superior characteristics compared with conventional methods in feed drive systems. However, with these methods, the control system design necessitates sufficient knowledge of control theory and an accurate dynamic model of the mechanism, including the machine base. These requirements limit its applications. Generally, it takes

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time and effort to obtain the characteristic information.

The nominal characteristic trajectory following (NCTF) control method has been proposed for conveniently realizing a high-precision position control system without any accurate dynamic model of the stage [17–19]. Its effectiveness has been demonstrated by the results applied to various mechanisms. However, vibration suppression methods are yet to be sufficiently investigated. No method has been demonstrated to suppress the adverse effects of vibrations caused by the machine base.

This paper presents a control system design for realizing high-speed and ultra-precision point-to-point positioning of a precision stage affected by unknown vibrations. The design does not necessitate knowledge of control theory or the machine's detailed dynamic characteristics. This applies to the stage installed on the machine base. The design aims to achieve high-speed and ultra-precision control of the stage's relative displacement with respect to the machine base. A linearized micro-dynamic model of the machine is used only to examine the control elements' effects on the vibration near the final target position. The design is performed in two steps: basic FB controller design and vibration suppression compensator design. The NCTF controller is used as the basic FB controller.

The remainder of this paper is organized as follows. Section 2 explains the ultra-precision linear motion stage as a control mechanism. In Section 3, the NCTF control system based on the ultra-precision linear motion stage is designed, and the residual vibration problem is investigated experimentally. In Section 4, the linearized micro-dynamic model of the experimental setup is presented. The effects of the compensating elements on the vibration characteristics are discussed using a micro-dynamic model, and an easy-to-use vibration suppression method is presented. Section 5 describes the procedure for adding and tuning the compensation elements for vibration suppression in the FB control system. The NCTF control system was used as the FB control system. Subsequently, the performance of the NCTF control system with the compensating elements was verified experimentally. Finally, Section 6 concludes this paper.

2. Ultra-precision linear motion stage on the machine base

Fig. 1 shows the schematic of a single linear motion stage on a machine base. This stage is used in this research. The linear motion stage to be controlled is installed on a machine base supported by four main columns. The table of the linear motion stage is guided by two linear ball guides and driven by a coreless linear motor. The working range is greater than 300 mm. In the experimental setup, two similar linear motion stages were installed side-by-side on the machine base in the translational direction. However, for simplicity and clarity of Fig. 1, one line motion stage used is illustrated on the machine base. A linear encoder with a resolution of 1 nm (HEIDENHAIN: LIC4117-440) was attached for table position measurement. An encoder was used as the feedback sensor. The sampling frequency for the measurements was 12 kHz.

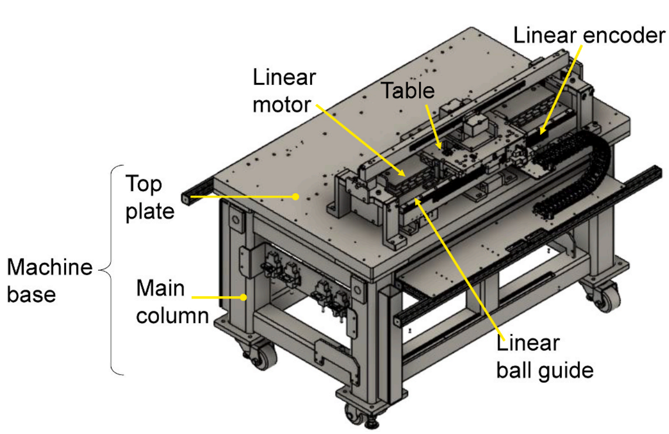

Fig. 1. Schematic of a single linear motion stage.

3. Vibration problem in high-speed positioning

3.1. NCTF control system

Fig. 2 shows the configuration of the NCTF control system. The controller has a nominal characteristic trajectory (NCT) expressed on the phase plane and a proportional plus integral (PI) compensator. NCT represents the desired approach characteristics of the target position and/or target motion trajectory. In this study, the NCT was used to achieve the pre-shaped target motion trajectory in point-to-point positioning. The role of the NCT is the same for both the pre-shaped and regular target motion trajectories. The NCT has proven useful and suitable for both point-to-point positioning as well as motion control in [20]. A simple open-loop response of the stage was used to determine the NCT without using the dynamic model. Fig. 3 shows the open-loop response of the linear motion stage to the square-wave input signal. In this study, the input signal was selected with an emphasis on high speed. Fig. 4 shows the measured trajectory using the open-loop experiment and its corresponding NCT. However, the measured trajectory contained parts of unwanted motions, including mechanism vibration and overshoot. After filtering the measured trajectory, the unwanted motion is removed from the filtered trajectory to construct the NCT, as shown in Fig. 4.

To reach and follow the target value, PI compensation constrains the operation of the controlled object to the NCT. The proportional gain K_p is empirically selected to be 0.6 times the maximum gain that is stable when using a P compensator. Subsequently, the integral gain K_I is adjusted. The following conditional freeze integrator is used to suppress the overshoot caused by the integral windup:

$$\Delta u_i = \begin{cases} 0, |u_o + u_i| > u_s \ \text{ and } \ e{\cdot}u_i \geq 0 \\ u_p, \text{otherwise} \end{cases}. \tag{1}$$

Here, u_p is the input to the integrator. u_o and u_i are the outputs of proportional element and integrator, respectively. Δu_i is the rate of change of u_i, and u_s is the maximum control signal. In this study, all compensators were discretized via the backward difference method and implemented in the experimental setup.

3.2. Vibration problem

The performance of the designed NCTF control system was experimentally examined using the target motion trajectory shown in Fig. 5. A pre-shaped trajectory was used to achieve the final target position at 300 mm. The acceleration waveform contained a triangular wave component. The maximum acceleration and velocity were 10 m/s^2 and 1 m/s, respectively. Fig. 6 shows the error response of the NCTF control system to the target trajectory in Fig. 5(c). Fig. 6(a) shows that the error is smaller than 80 μm, thereby proving that the NCTF control system worked effectively. Fig. 6(b) shows an enlarged view of the error

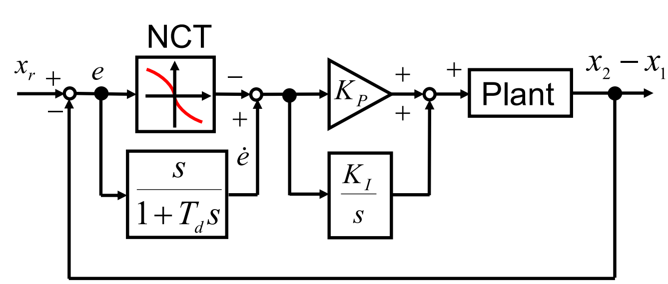

Fig. 2. Configuration of the NCTF control system.

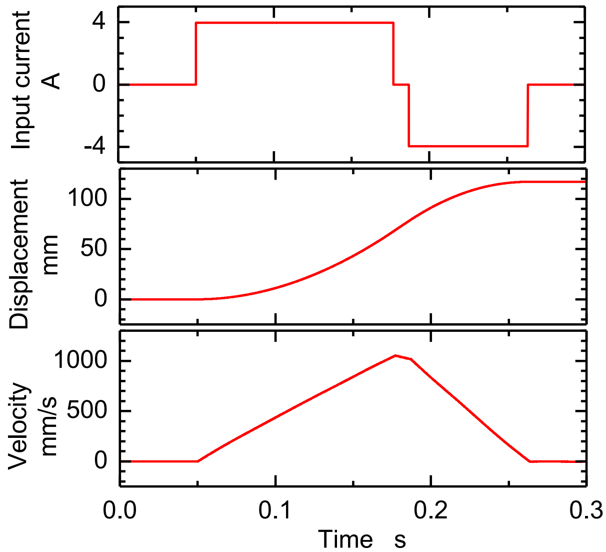

Fig. 3. Open-loop response of the linear motion stage to the square wave input signal.

response near the final target position. The fundamental role of the control system is to achieve high-speed and ultra-precision point-to-point positioning and not to focus on tracking motion control. Therefore, it is important to suppress the residual vibration with an amplitude of 50 nm and frequency of approximately 20 Hz. It is considered that this vibration is generated in the traveling direction by the reaction force that is applied to the top plate of the machine table when the table is driven. A vibration of approximately 20 Hz was also observed when the two linear motion stages of the experimental setup were connected and driven. In addition to the frequency of 20 Hz, the vibration contains a component with a smaller amplitude at a higher frequency. The high-frequency vibration component is considered to be related to the characteristics of the linear motion stage. Subsequently, compensators and their tuning processes rapidly suppress the vibration.

4. Practical vibration suppression method

4.1. Dynamic model

In this study, a dynamic model is used for the vibration suppression analysis of the table's relative position with respect to the machine base. The linear motion stage is driven by a stroke of 300 mm for evaluation. To improve the positioning accuracy and shorten the positioning time, it is important to rapidly suppress the residual micro-vibration near the target relative position. Fig. 7 shows a micro-dynamic model of the linear motion stage on the machine base. A description of the model parameters is presented in Table 1. Here, c_3 is the coefficient of viscous friction force that is independent of motion range. It can be determined from the relationship between the driving force and the table velocity in macroscopic motion range. In the fine motion region, the friction characteristics of rolling guides exhibit spring-like characteristics [21–24]. The static friction element behaves like the combination of a spring and viscous friction elements in the fine motion region. Here, c_2 represents the coefficient of the viscous friction element caused only in the fine motion region. Thus, the value of c_3 previously determined is independent of that of c_2. The impacts of c_2 and c_3 are the same, and the sum of their value can be determined from the vibration characteristics in the fine motion region. Consequently, the value of c_2 can be determined based on the difference between the sum and the value of c_3. The error characteristics are expressed by linear elements.

In Section 3, the effectiveness of the controller structure for vibration suppression is examined using the characteristics related to the relative position (x_2 - x_1). The transfer function $G_{tp}(s)$, from the motor driving force p to the relative position (x_2 - x_1), is expressed as a function of the complex number s as follows:

$$G_{tp}(s) = \frac{X_2 - X_1}{P} = \frac{G_{m1}(s) + m_2 s^2}{G_{m1}(s)G_{m2}(s) + m_2 s^2 G_{e2}(s)}, \tag{2}$$

$$\left.\begin{aligned} G_{m1}(s) &= m_1 s^2 + c_1 s + k_1 \\ G_{m2}(s) &= m_2 s^2 + (c_2 + c_3)s + k_2 \\ G_{e2}(s) &= (c_2 + c_3)s + k_2 \end{aligned}\right\}. \tag{3}$$

The transfer function $G_{td}(s)$ from the disturbance force d applied to the machine base to the relative position (x_2 - x_1) is as follows:

$$G_{td}(s) = \frac{X_2 - X_1}{D} = \frac{-m_2 s^2}{G_{m1}(s)G_{m2}(s) + m_2 s^2 G_{e2}(s)}. \tag{4}$$

Residual vibration is considered to occur when a disturbance force is applied to the machine base. The vibration suppression ability is comprehended using Eq. (4).

4.2. Effect of compensators for vibration suppression

When residual vibration deteriorates position accuracy, vibration can be suppressed effectively by increasing the FB gain for the vibration component and appropriately advancing the phase of the FB signal. In this section, two types of compensators with these characteristics are

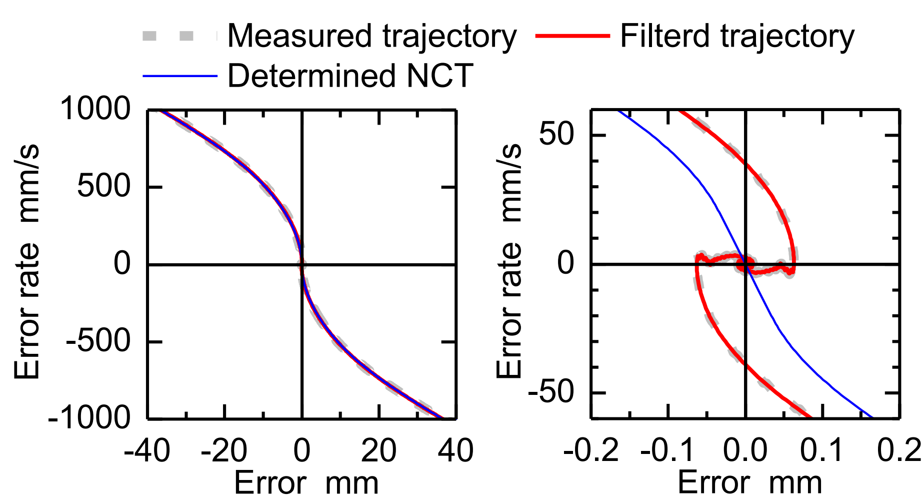

Fig. 4. Measured trajectory and NCT determined on the phase plane.

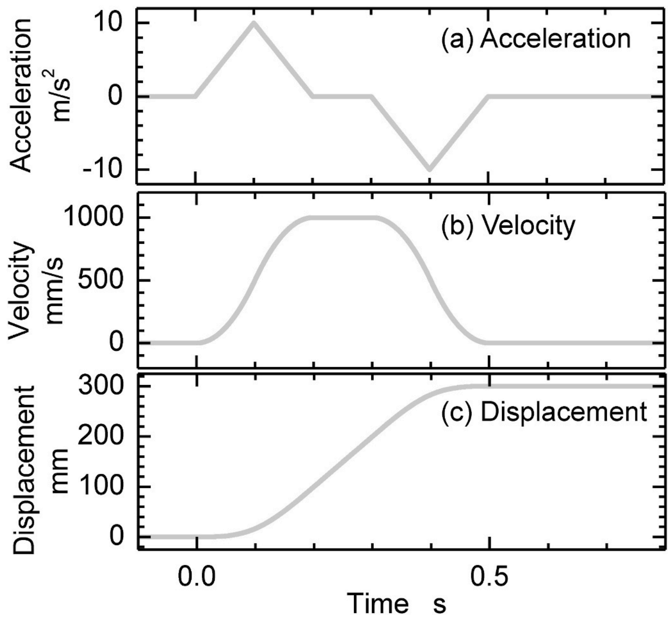

Fig. 5. Pre-shaped target motion trajectory (1).

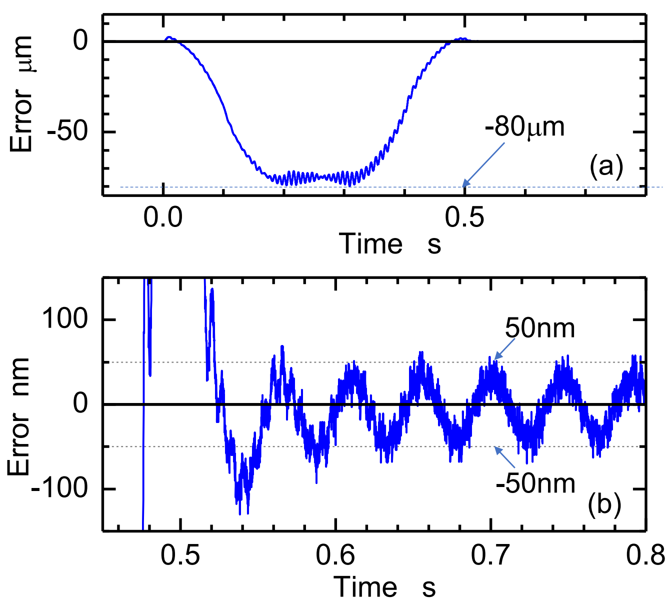

Fig. 6. Error response of the NCTF control system.

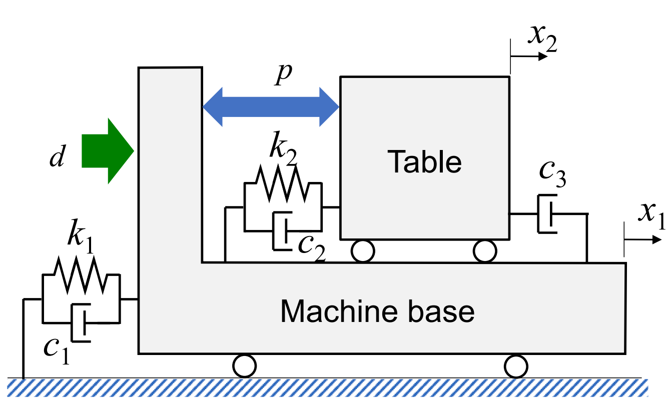

Fig. 7. Micro-dynamic model of the linear motion stage on the machine base.

Table 1
Model parameters.

Symbol	Description	Value
m_1	Mass of the machine base	3.26×10^2 (kg)
m_2	Mass of the table	1.259×10 (kg)
k_1	Stiffness of machine-base support	5.24×10^6 (N/m)
c_1	Viscous friction coefficient of the machine-base support	5.7×10^3 (Ns/m)
k_2	Stiffness in the micro-dynamic characteristic	1.36×10^5 (N/m)
c_2	Viscous friction coefficient of the micro-dynamic characteristic	8.75×10^2 (Ns/m)
c_3	Viscous friction coefficient in the stage	1.5×10^3 (Ns/m)
P	Motor driving force	
D	Disturbance force	
x_1	Position of the table	
x_2	Position of the machine base	

presented, and their effects are examined using the linear dynamical model presented in Section 3.2.

4.2.1. Compensator that increases the system stiffness at a specific frequency

To suppress a specific frequency component, it is effective to add a bandpass filter (BPF) in parallel [25]. A block diagram of the control system with the BPF is shown in Fig. 8. Compensator $B(s)$ is expressed as

$$B(s) = \frac{s^2 + 2\zeta_1\omega s + \omega^2}{s^2 + 2\zeta_2\omega s + \omega^2} = 1 + \frac{2(\zeta_1 - 2\zeta_2)\omega s}{s^2 + 2\zeta_2\omega s + \omega^2}\,(\zeta_1 > \zeta_2). \tag{5}$$

The symbol ω is the resonance angular frequency that can be easily determined from the time–response characteristics. The effect of the compensator $B(s)$ on the frequency characteristics, related to the relative position, is shown in Fig. 9. In the figure, ω is determined to suppress the machine base vibration. The transfer function $M_{trB}(s)$, which shows the target value followability of the control system, is expressed as follows:

$$M_{trB}(s) = \frac{X_2 - X_1}{R} = \frac{G_{TF}(s)C_{PI}(s)B(s)G_{tp}(s)}{1 + G_{TF}(s)C_{PI}(s)B(s)G_{tp}(s)}, \tag{6}$$

$$\left.\begin{aligned} G_{TF}(s) &= m + \frac{s}{1 + T_d s} \\ C_{PI}(s) &= K_p + \frac{K_I}{s} \end{aligned}\right\}, \tag{7}$$

where m is the slope near the origin of the NCT. The transfer function $M_{tdB}(s)$ that shows the disturbance response characteristics of the control system is expressed as follows:

$$M_{tdB}(s) = \frac{X_2 - X_1}{D} = \frac{G_{td}(s)}{1 + G_{TF}(s)C_{PI}(s)B(s)G_{tp}(s)}. \tag{8}$$

Fig. 9(a) shows the disturbance response characteristics expressed in $M_{tdB}(s)$. A small gain indicates that the system has a strong ability to suppress residual vibrations. It can be observed that a smaller ζ_2 reduces the relative position vibration caused by the disturbance force with an angular frequency of ω. Therefore, it is effective in reducing the residual

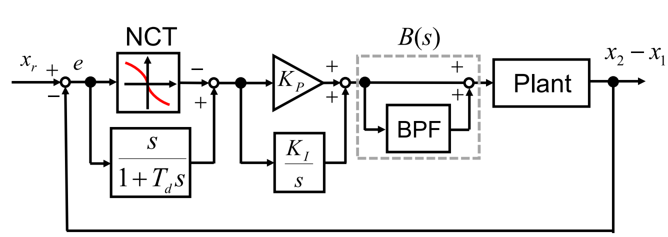

Fig. 8. NCTF control system with BPF.

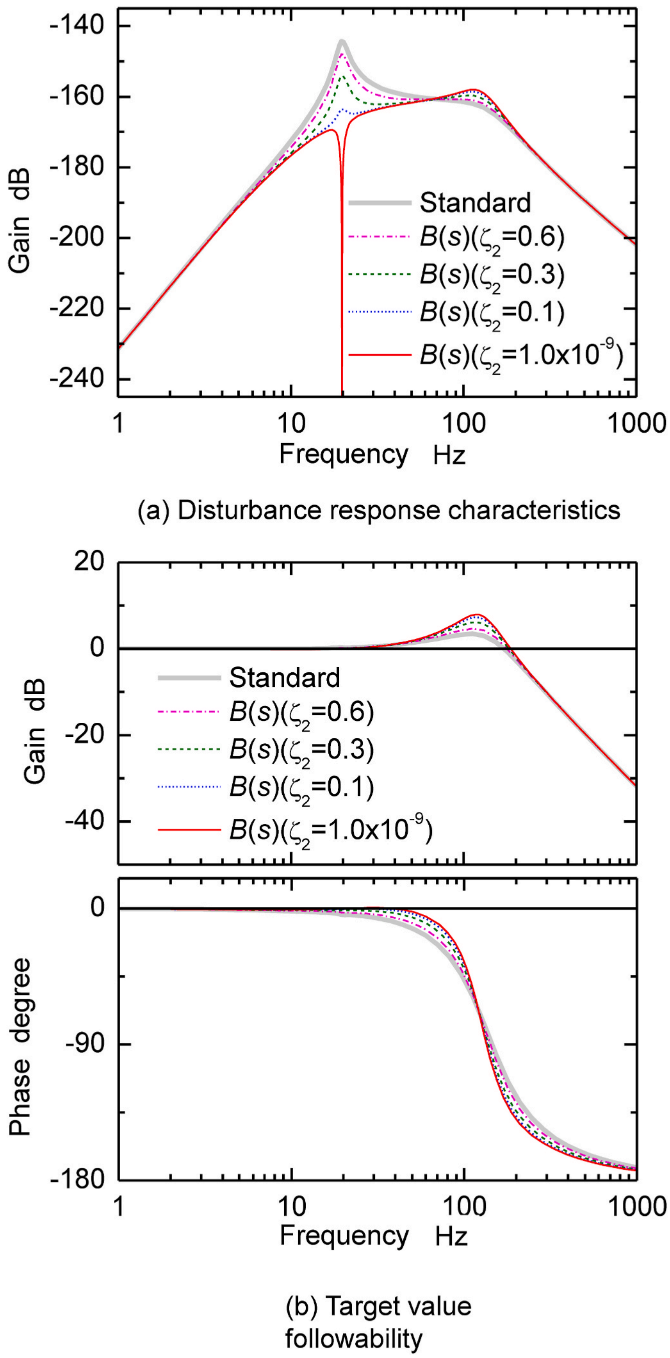

Fig. 9. Effect of compensator $B(s)$ on the frequency characteristics related to the relative position (x_2-x_1).

vibration. Nonetheless, increase in gain in the high-frequency band and degradation of disturbance suppression characteristics are problematic. Consequently, resonant vibration of the stage can be induced. The target value followability obtained using $M_{trB}(s)$ is shown in Fig. 9(b). The figure indicates that $B(s)$ with a small ζ_2 can increase the overshoot during step input. The risk can be avoided by activating $B(s)$ after the stage table is sufficiently close to the target value.

$B(s)$ with a small ζ_2, designed for high-frequency vibration, can also suppress the vibration caused in the high-frequency band. However, this deteriorates the characteristics of the higher-frequency bands. Fig. 10 shows the disturbance response characteristics of the NCTF control system, with improved robustness against disturbance forces, at a specific high frequency using $B(s)$. The gain is reduced at a particular

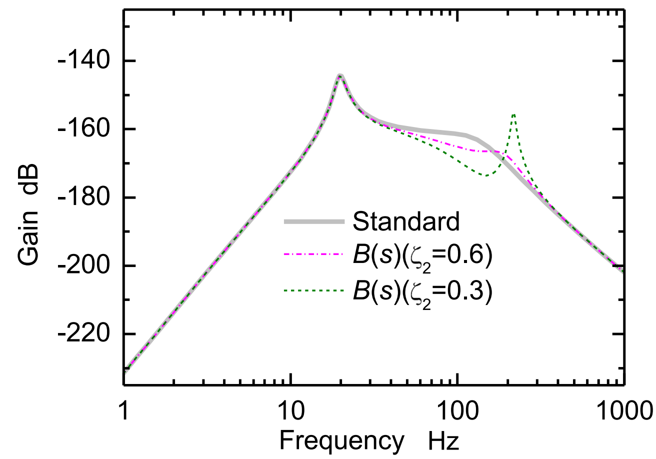

Fig. 10. Disturbance response characteristics of the NCTF control system with high natural frequency $B(s)$.

frequency, but the disturbance response characteristics are degraded in the nearby high-frequency band. $B(s)$ also degrades the target value followability of the control system in the band. Therefore, it is desirable to use $B(s)$ with another compensator to suppress the adverse effects.

4.2.2. Derivative compensator

Fig. 11 shows a block diagram of the NCTF control system with a derivative compensator $F(s)$ ($= K_c s$). Derivative compensators are often used to suppress vibrations. In this control system, the derivative compensator operates, such that the table may stop near the target value. Fig. 12 shows the frequency characteristics of the disturbance response of the control system, as shown in Fig. 11. The gain effectively decreases at a frequency of approximately 100 Hz. No bulge around 165 Hz is observed. This indicates that the compensator can improve the vibration-suppression characteristics in the high-frequency range. However, the suppression effect decreases as it approaches the resonance peak.

As aforementioned, the derivative compensator has characteristics different from those of the compensators described in Section 4.2.1 in that the latter can increase the system stiffness at a specific frequency and compensate for the shortcomings of the individual differentiators. Therefore, the combination of their compensators can effectively suppress the vibrations generated in a system with a resonance point in the low-frequency region. Fig. 13 shows an example of the response characteristics of control systems with two compensators. The figure shows the effect of the derivative gain K_c in the control system, in which the resonance peak is suppressed at $B(s)$ with $\zeta_2 = 0.1$. As shown in Fig. 12, the derivative compensator removes the bulge at approximately 165 Hz. In the frequency band near and below the resonance frequency, the gain decreases. However, it has no effect on the resonance peak gain near 20 Hz. Furthermore, the gain increases and the disturbance suppression

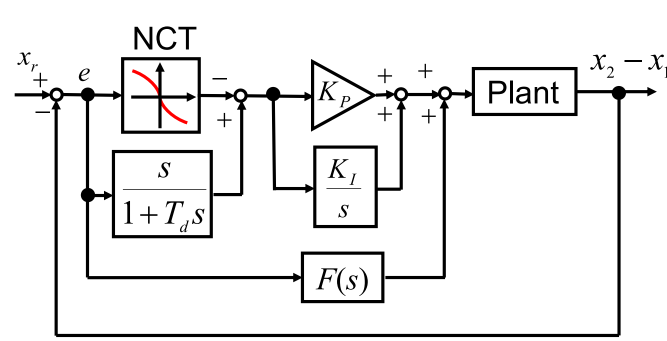

Fig. 11. NCTF control system with a derivative compensator $F(s)$ ($= K_c s$).

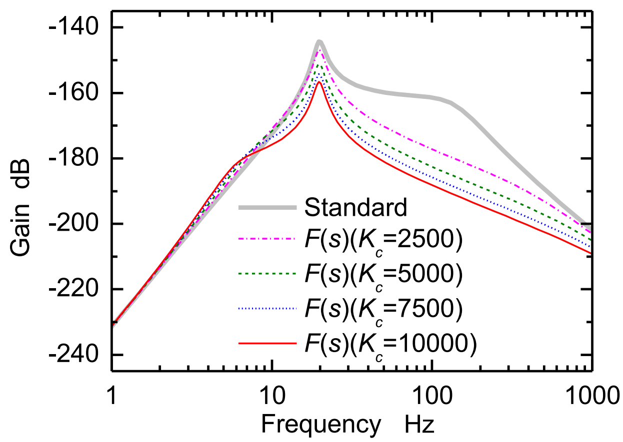

Fig. 12. Disturbance response characteristics of the NCTF control system with $F(s)$.

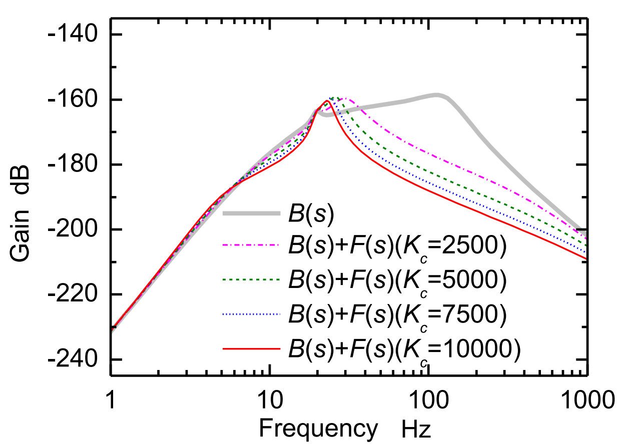

Fig. 13. Example of the response characteristics of control systems with the two compensators.

characteristics degrade near the resonance frequency. The results indicated that the combined use of the two compensators is effective and necessary for vibration suppression.

5. Ultra-precision position control system with vibration suppression compensators

This section introduces the design of the control system, including the two types of compensators described in Section 4, and demonstrates their effects obtained through experiments.

5.1. Design of vibration suppression compensators

After the NCTF control system was designed, two compensators were added and tuned. The tuning procedure and specific work for the experimental linear motion stage are as follows. In this section, the characteristics of the control system are discussed using the responses of the control systems to the trajectory shown in Fig. 5(c). The control system designed in this study is shown in Fig. 14. The compensators were activated after the target motion trajectory reached its final value. This prevented $B(s)$ from affecting the target value followability and ensured the combination suppressed the residual vibration effectively.

Step 1: Determining problematic vibration frequency f_s

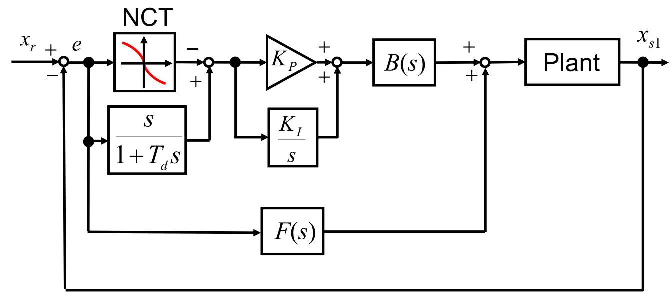

Fig. 14. Vibration suppression of the NCTF control system with the two compensators.

f_s was determined from the waveform near the target position obtained from the NCTF control system response to the trajectory given in Fig. 5(c) f_s of the experimental setup is 19.78 Hz and caused by the machine base. The angular frequency ω in Eq. (5) was determined to be $2\pi f_s$.

Step 2: Tuning ζ_2 in $B(s)$

For simplicity, ζ_1 was fixed at 0.9, and only ζ_2 was tuned. There was only one adjustment parameter in this step, and it was easy to determine the most effective ζ_2 by reducing ζ_2. Fig. 15 shows its effect on the residual error. As ζ_2 decreased, the vibration at approximately 20 Hz decreased, but the high-frequency vibration increased. Fig. 16 shows the FFT analysis results over 1 s after reaching the target position. A smaller ζ_2 effectively reduced the low-frequency vibration component, at approximately 20 Hz, but significantly increased the high-frequency component at approximately 165 Hz. The cause of the high-frequency vibration was unclear, but it was considered to be related to the low stiffness element of the stage. This tendency is consistent with the analysis of the dynamic model shown in Fig. 7. In this step, $\zeta_2 = 0.3$ was selected with an emphasis on the effectiveness in the low-frequency range. This value is the smallest in the stable range.

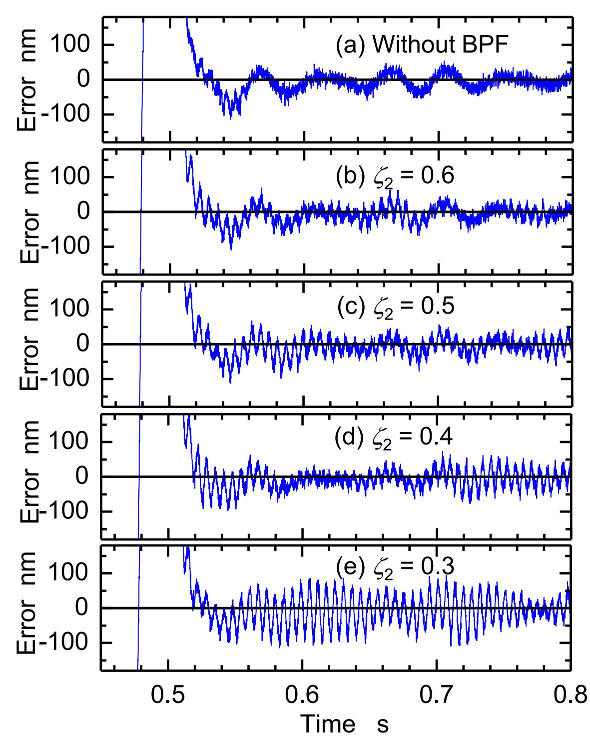

Fig. 15. Effect of ζ_2 on the residual error.

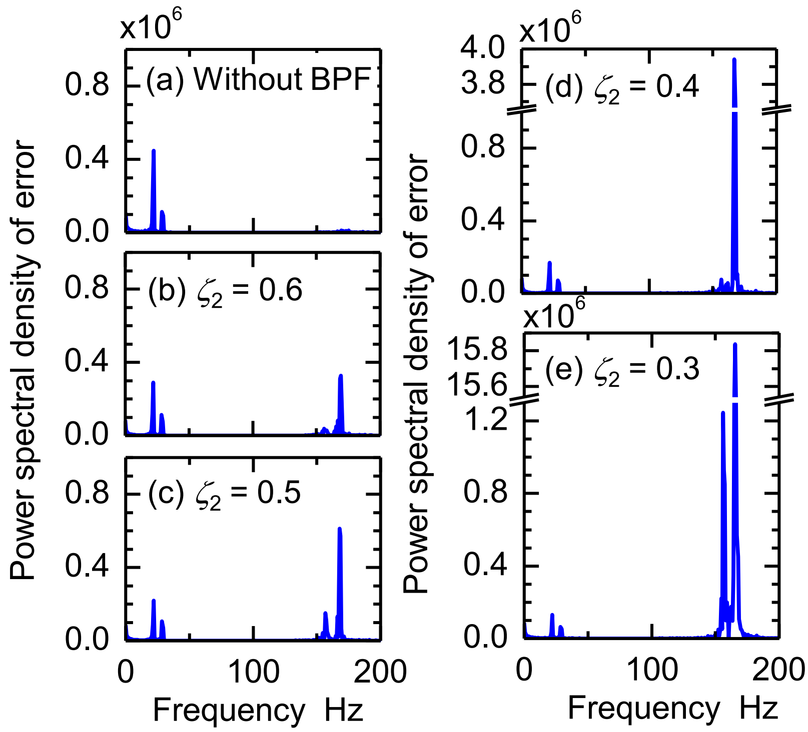

Fig. 16. Effect of ζ_2 on the power spectral density of the error.

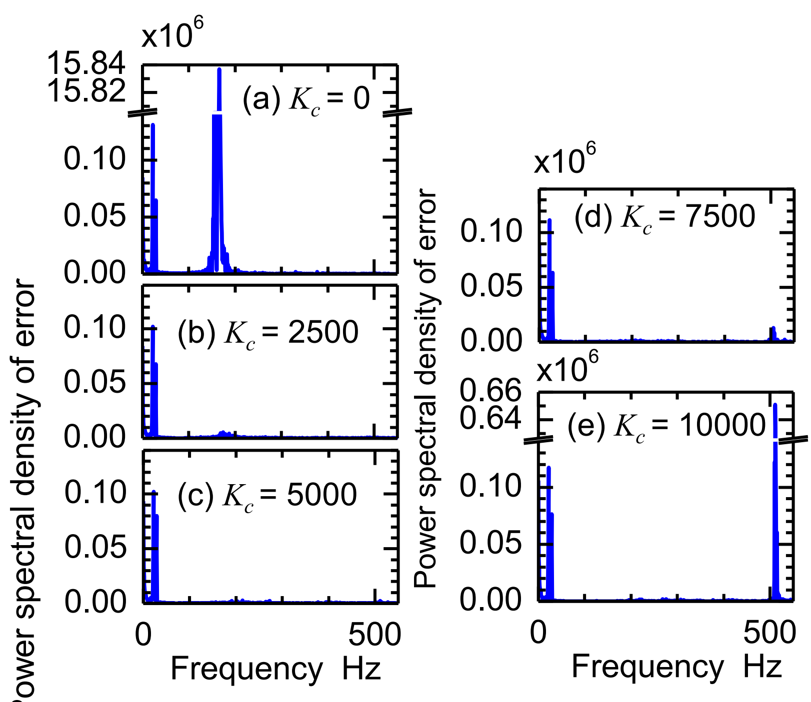

Fig. 17. Effect of derivative element on the power spectral density of the error ($\zeta_2 = 0.3$).

Step 3: Tuning K_c of $F(s)$

At $\zeta_2 = 0.3$, only the derivative gain K_c was tuned. The effect of its gain, on the error's frequency analysis results, is shown in Fig. 17. With $K_c \leq 5000$, the high-frequency oscillation induced by $B(s)$ can be effectively suppressed. However, a larger K_c increased the oscillation above 500 Hz. Therefore, $K_c = 5000$ was selected for this study. The error response of the control system determined in this step is illustrated in Fig. 18.

Step 4: Retuning ζ_2 in $B(s)$

This step was performed when the problematic vibration needed further suppression. The procedure was the same as in Step 2. Vibrations with frequencies higher than 100 Hz limited the increase in ζ_2 in Step 3. However, as shown in Fig. 17, the power-spectral density of the error around 20 Hz was considerably larger than that at frequencies higher than 100 Hz. Thus, ζ_2 was returned to further suppress the resonance vibration at approximately 20 Hz. Fig. 19 shows the effect of $B(s)$ on the error's power spectral density. It can be observed that the vibration near 20 Hz can be further suppressed by reducing ζ_2 in $B(s)$. The vibration near 165 Hz hardly increases, even when ζ_2 is reduced. Therefore, the tuning was completed with $\zeta_2 = 0$.

5.2. Control-system performance

The responses of the control system determined in Sections 3.1 and 5.1 were measured and examined. Fig. 20 shows the error response of the control system to the target motion trajectory shown in Fig. 5. As shown in Fig. 6, the residual error of the conventional NCTF control system was approximately 50 nm but did not converge below 50 nm. However, the proposed control system converged within 0.08 s after the target value reached the final value. In addition, the vibration at approximately 20 Hz was rapidly attenuated.

The performance of the proposed control system was examined using the target motion trajectory shown in Fig. 21. The pre-shaped trajectory was generated based on the sinusoidal wave acceleration with the aim of achieving the final target position. The maximum acceleration and velocity in Fig. 21 are the same as those in Fig. 5. The final target displacement is also the same as that in Fig. 18, but the change rate in the target acceleration is smaller than that in Fig. 5. A comparison between Figs. 20 and 22 shows that the reduction in the change rate is effective in reducing the residual vibration, even in the control system with vibration suppression compensators considered in this study. In the response of such a control system, the vibration caused by the machine base is extremely small. Conversely, this vibration is evident in the response of the conventional NCTF control system. The results indicated that the proposed compensators have a significant effect on vibration suppression.

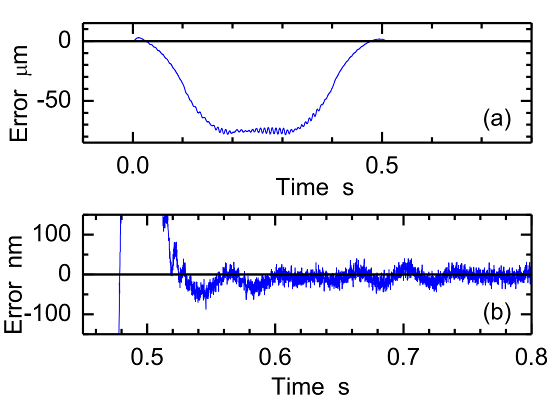

Fig. 18. Error response of the control system determined in Step 3 ($\zeta_2 = 0.3$, $K_c = 5000$).

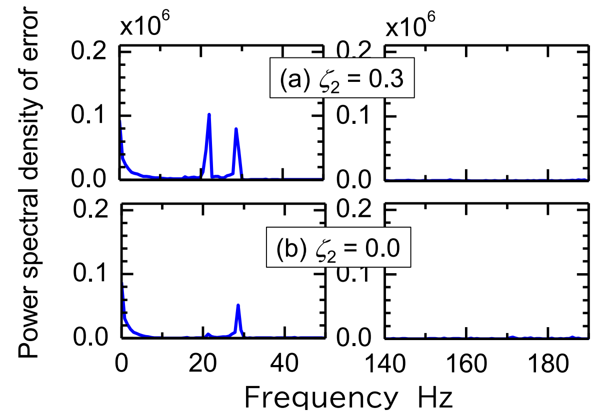

Fig. 19. Effect of Step 4 on the power spectral density of the error.

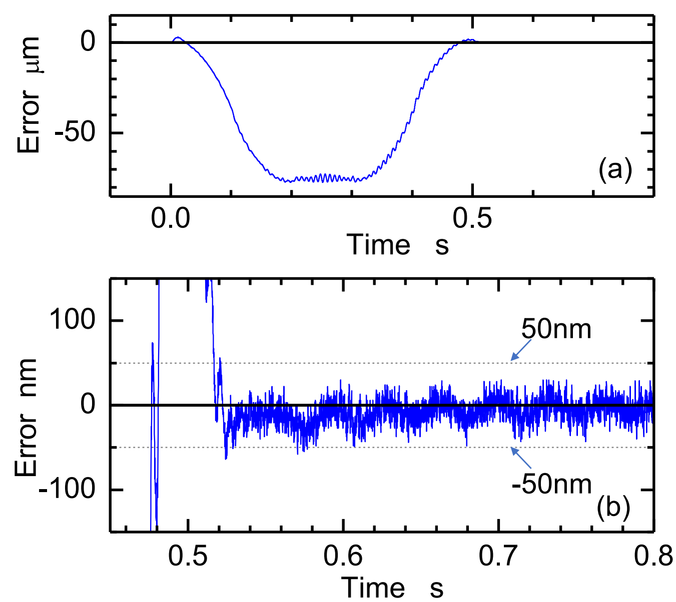

Fig. 20. Error response of the control system to the pre-shaped target motion trajectory (1).

6. Conclusion

The high-speed drive of the stages causes machine bases to vibrate, which usually deteriorates position accuracy. To overcome this problem, two types of simple but effective vibration suppression compensators were examined. The design procedure and performance of the control system with these compensators were demonstrated. Each compensator featured one tuning parameter and complementary characteristics for the other. This feature was utilized for tuning the parameter. To simplify the control-system design, these compensators were incorporated into the NCTF control system.

The high-precision control of a mechanism, with various friction and saturation characteristics, can be easily realized in the NCTF control

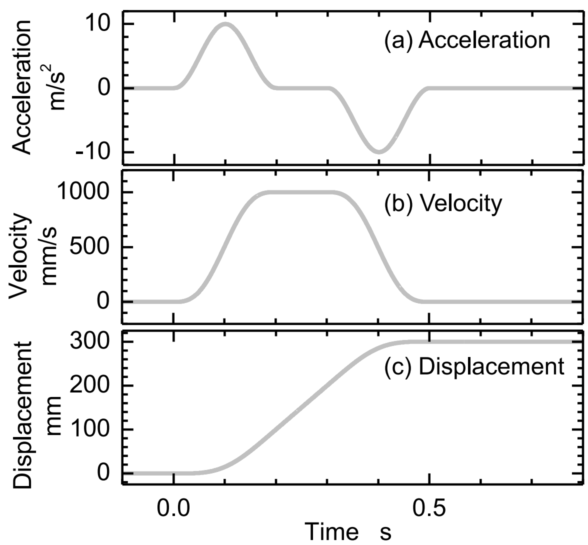

Fig. 21. Pre-shaped target motion trajectory (2).

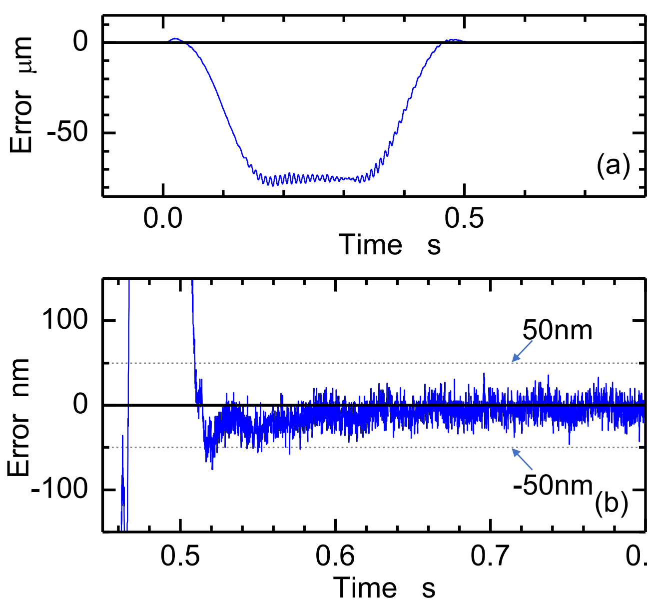

Fig. 22. Error response of the control system to the pre-shaped target motion trajectory (2).

system. It does not require accurate mechanism information and can be adjusted independent of the compensators. Two types of target motion trajectories were applied to the designed control system for performance evaluation. The experimental results demonstrated the effectiveness of the compensators and high performance of the control system. The results are expected to be effective for existing control systems and more complicated vibration systems with unknown characteristics. Future work will determine a controller design procedure for ultra-precision positioning of a more complicated and unclear vibration system, based on these results.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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